

Restaurant & Food Inventory Database Management System

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INTRODUCTION

Did you know?

Handles hundreds of ingredients and orders daily.



A Restaurant & Food Inventory Management System is a database-driven software application that helps restaurants manage:

- Food ingredients
- Inventory stock
- Suppliers
- Customer orders
- Menu items
- Purchase records
- Sales records

Instead of maintaining records manually, the system stores everything in a centralized database, making restaurant operations faster, more accurate, and more efficient.

Proposed Solution

Our system will provide:

- ✓ Inventory Management
- ✓ Order Management
- ✓ Supplier Management
- ✓ Menu Management
- ✓ Employee Management
- ✓ Sales Tracking
- ✓ Purchase Tracking
- ✓ Automatic Stock Updates
- ✓ Report Generation

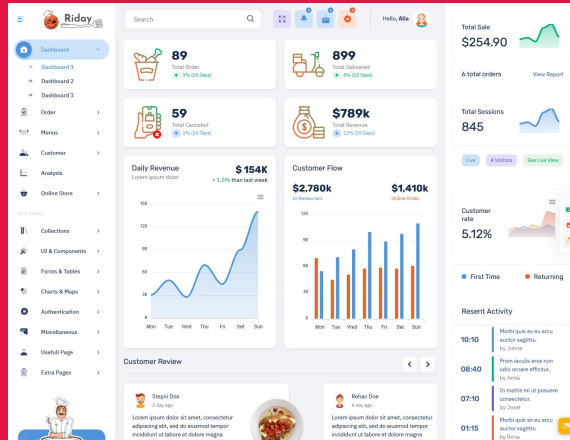


Every customer order automatically updates the inventory database. Managers always know current stock levels without checking manually.

Problems with Traditional Management

Restaurants often face:

- Manual record keeping
- Lost inventory records
- Food wastage
- Stock shortages
- Duplicate data
- Slow report generation
- Human errors
- Difficulty tracking suppliers



Digital Dashboard



Manual Paperwork

Without a proper database, restaurants struggle to keep track of inventory and sales. This causes unnecessary costs and poor customer service

Objectives

Primary Objectives

- Build a centralized restaurant database
- Reduce food waste
- Improve inventory accuracy
- Manage suppliers efficiently
- Track customer orders
- Generate reports quickly



Secondary Objectives

- Improve decision making
- Save employee time
- Reduce operational cost
- Increase restaurant productivity

Features of the system

INVENTORY MODULE

Add stock

Update stock

Remove expired stock



ORDER MODULE

Customer orders

Order history

Bill generation



SUPPLIER AND EMPLOYEE MODULE

Supplier information

Purchase history

Employee records

Roles



REPORTS

Daily Sales

Monthly Sales

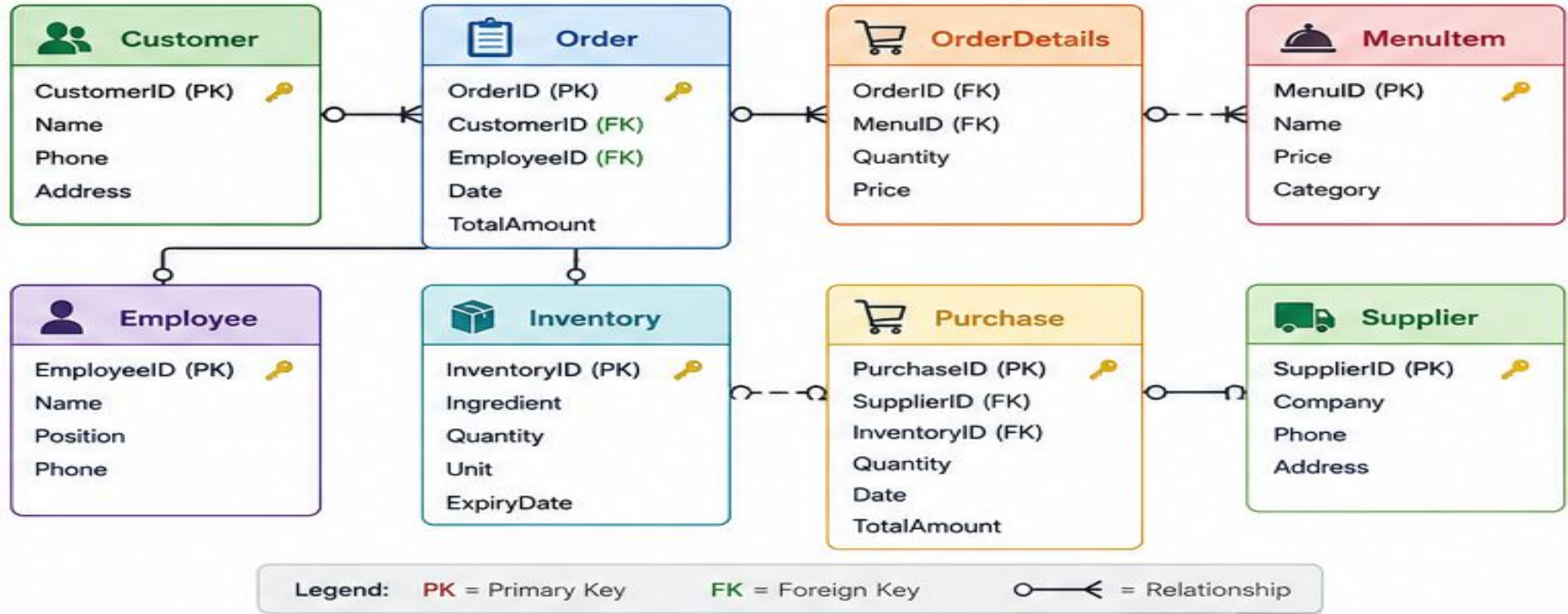
Inventory Status

Low Stock Alert



DATABASE DESIGN

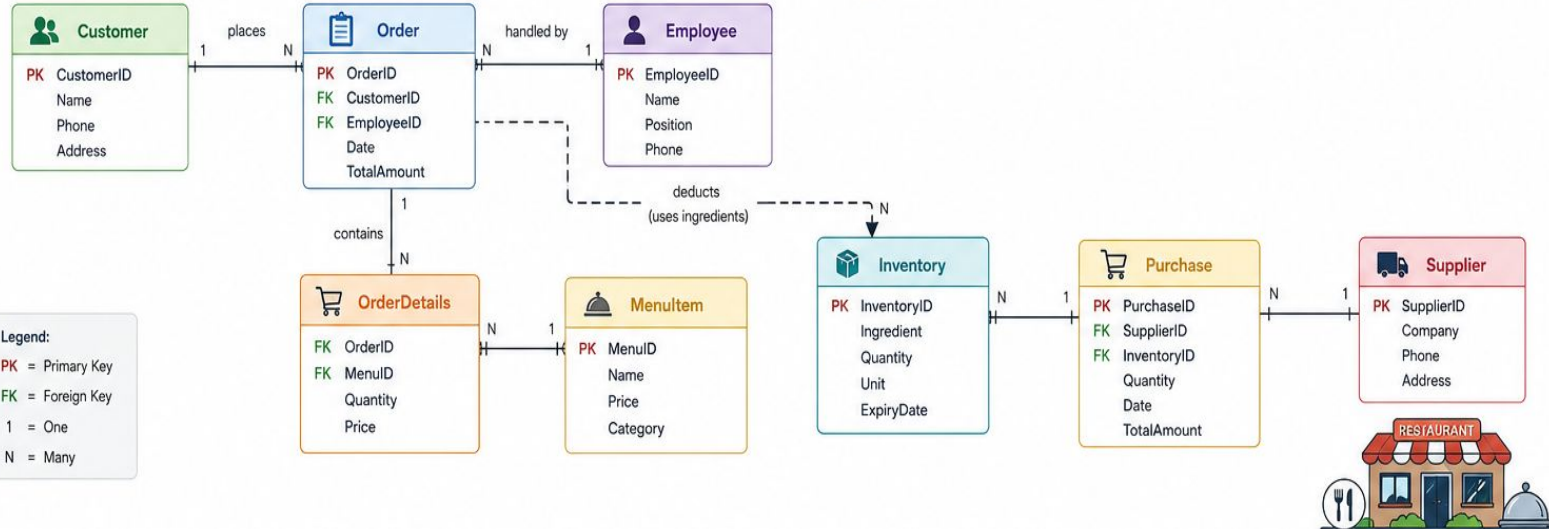
MAIN DATABASE TABLES



Every table has a primary key, and foreign keys connect related tables.

Entity Relationship Diagram(ERD)

Slide 7 – Entity Relationship Diagram (ERD)



One Customer →
Many Orders

One Order →
Many Menu
Items

One Supplier →
Many Purchases

Inventory
updates after
each order.



NORMALIZATION

Normalization improves database performance and prevents duplicate information

1NF

First Normal Form

No repeating groups.

2NF

Second Normal Form

Every non-key attribute depends on the whole primary key.

3NF

Third Normal Form

No transitive dependency.

DBMS Concepts Used

01

Primary Key

Unique Identification

02

Foreign Key

Connects Tables

03

Constraints

Not Null

Unique

Check

05

Joins

Inner Join
Left Join

04

SQL Operations

INSERT
UPDATE
DELETE
SELECT

ADVANTAGES

Benefits

- ✓ Faster inventory management
- ✓ Accurate stock records
- ✓ Reduced food waste
- ✓ Better supplier management
- ✓ Improved customer service
- ✓ Easy report generation
- ✓ Secure data storage
- ✓ Better business decisions

Automation improves restaurant efficiency.

TECHNOLOGY STACK

FRONTEND

HTML

CSS

JAVASCRIPT



BACKEND

PHP/[node.js](#)



DATABASE

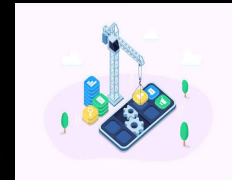
MySQL



TOOLS

XAMPP

phpMyAdmin
Visual Studio
Code



Even if implementation isn't complete, this as our proposed stack.

Future Scope

- Online food ordering
- QR Code menu
- Mobile application
- AI demand prediction
- Barcode scanner
- SMS stock alerts
- Cloud database
- Employee attendance

The project can be expanded into a real commercial system.

CONCLUSION

The Restaurant & Food Inventory Database Management System provides an efficient way to manage restaurant operations using a relational database.

It helps:

- Improve efficiency
- Reduce manual work
- Prevent inventory loss
- Increase productivity
- Support better business decisions

